Week 2 - Manufacturing & AI: Use-Case & Solution-Flow

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Part A: Pain-Point Map

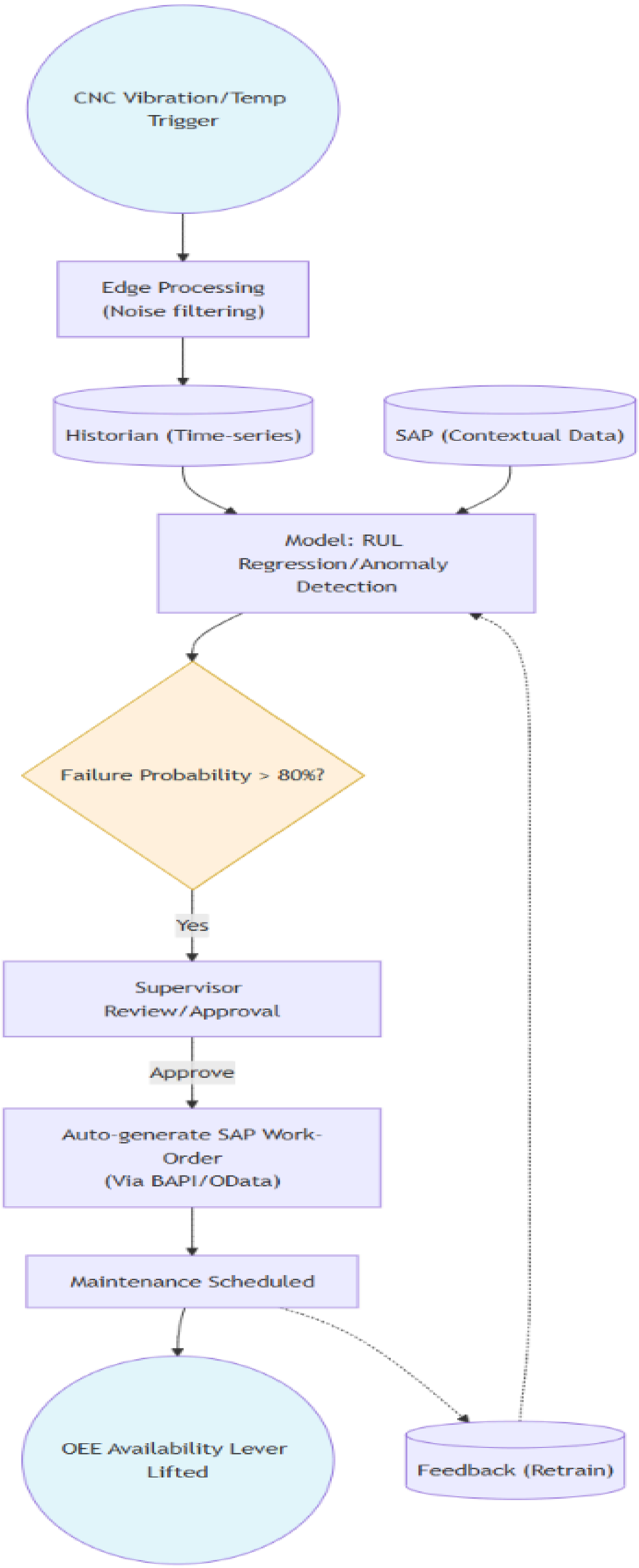
Pain Point	Business Impact	AI-Solvable?	Use Case Family	OEE Lever
Unplanned CNC stops	Significant downtime; production loss.	Yes	Predictive Maintenance (PdM)	Availability
Rising scrap rate	Increased material costs/waste.	Yes	Computer Vision (Quality)	Quality
Spreadsheet planning	Inefficient production scheduling.	No	Process/Change Mgmt	Performance

For Unplanned CNC Stops:
AI solves this by using Predictive Maintenance (PdM) to analyze real-time vibration data, allowing it to predict a breakdown before it happens so that repairs can be scheduled without causing unplanned downtime.

For Rising Scrap Rates:
AI solves this using Computer Vision, where cameras scan parts during production to detect surface defects instantly, enabling the system to stop the line or divert the item before it becomes scrap.

For Spreadsheet Planning:
This is not AI-solvable because it is a process and change-management problem; it requires replacing manual scheduling software with an integrated digital system rather than using a machine learning model

Part B: Solution Flowchart



Flow Logic:

1. **Signal:** Raw vibration and temperature data from CNC sensors.
2. **Connectivity:** Edge device filters high-frequency noise before data hits the platform.
3. **Context:** Data is stored in a Historian and contextualized with SAP.
4. **Model:** RUL Regression model predicts when the tool/bearing will fail.
5. **Decision:** If probability of failure > 80%, trigger maintenance process.
6. **Action:** Auto-generate a Work Order in SAP using BAPI/OData.
7. **Human in the Loop:** Maintenance lead reviews the alert/order and confirms.
8. **Feedback:** After repair, technician logs "actual failure cause" to retrain the model.

Part C: Justification Note

Why this technique?
We chose Predictive Maintenance (PdM) because CNC failure is a classic "time-series" problem. The machine doesn't just stop; it gives off warning signs (vibrations) that change over time. AI is the only way to catch these patterns before they cause a full shutdown.

Data & Integration:
We avoid reading the SAP database directly (which is slow and risky). Instead, we use BAPI or OData services, which are standard, safe ways for external systems to "talk" to SAP. This ensures our AI model integrates seamlessly without crashing the ERP.

Top 3 Risks:

1. **False Alarms:** AI flagging a healthy machine wastes time.
   Mitigation: We use a "Human-in-the-loop" step, the technician always confirms the alert before stopping the line.
2. **Master Data Errors:** If SAP doesn't know which sensor belongs to which machine, the model fails.
   Mitigation: Perform a data audit before launch.
3. **Change Resistance:** Workers may ignore the AI.
   Mitigation: Involve the maintenance team in setting the "Failure Threshold" for the MVP.

MVP: Our MVP is limited to one CNC machine and one specific part. We prove it works there, then scale to the rest of the line.